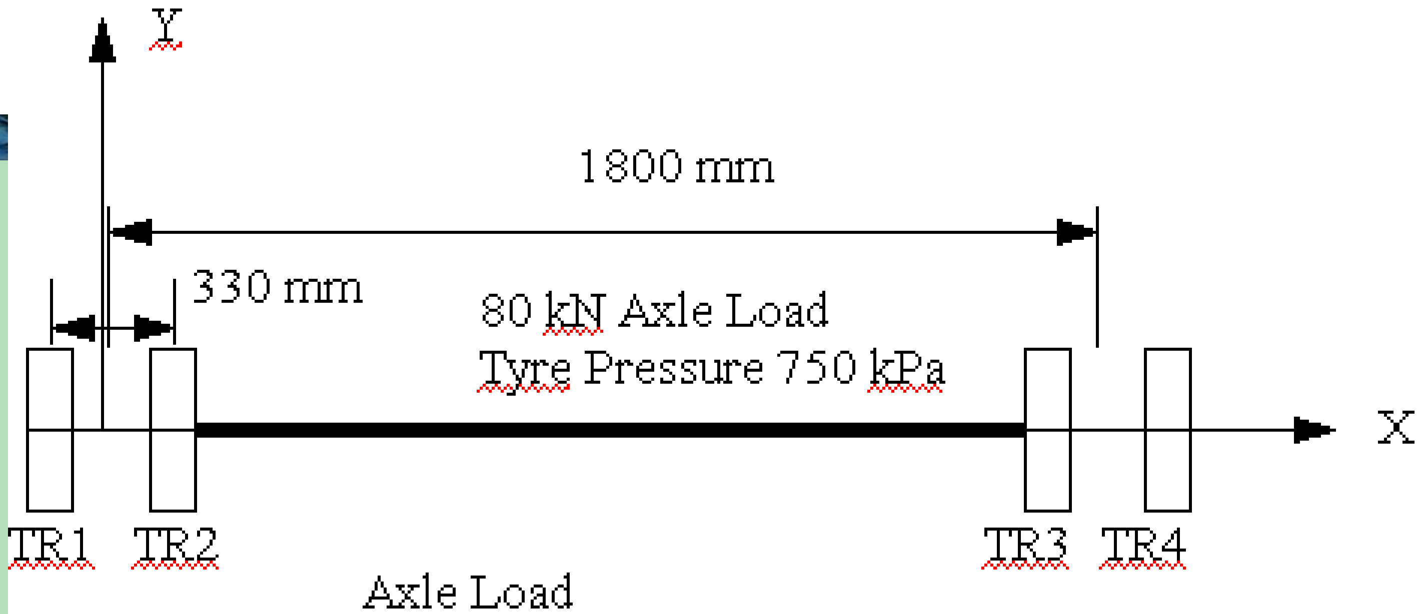
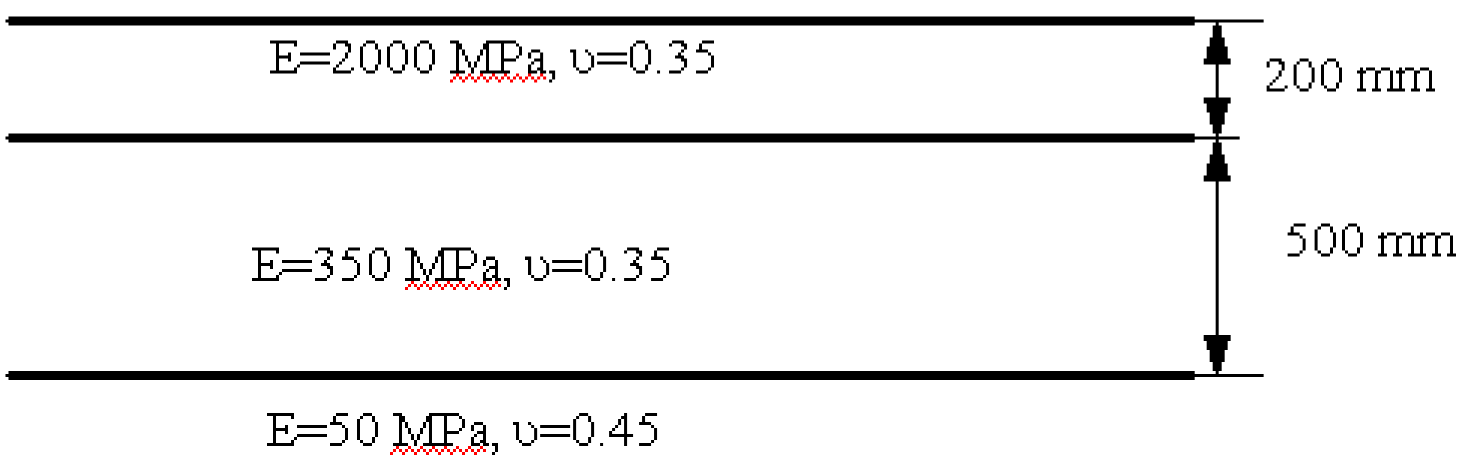
Tutorial 3
Circly Analysis

Mofreh Saleh, PhD, P.E., F.ASCE.

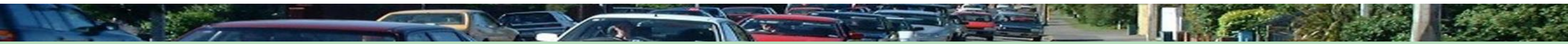

Y
1800 mm
330 mm
80 kN Axle Load
Tyre Pressure 750 kPa
X
TR1
TR2
TR3
TR4
Axle Load


E=2000 MPa, υ=0.35
200 mm
E=350 MPa, υ=0.35
500 mm
E=50 MPa, υ=0.45

Case 1: Cross Anisotropic Modelling for unbound materials

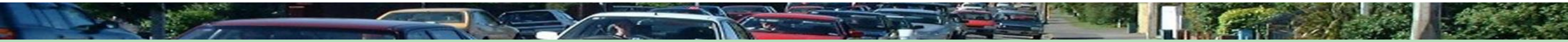
- Model the basecourse as cross anisotropic with no sublayering and the subgrade as cross anisotropic materials and determine:
- The maximum Surface Deflection and its Location
- The maximum tensile strain at the bottom of the asphalt layer (ε_{xx} or ε_{yy})
- The Maximum Compressive strain on the top of the subgrade (ε_{zz})

Case 2: Cross Anisotropic and sublayering Modelling for unbound materials



- Model the basecourse as cross anisotropic with sublayering and the subgrade as cross anisotropic materials and determine:
- The maximum Surface Deflection and its Location
- The maximum tensile strain at the bottom of the asphalt layer (ε_{xx} or ε_{yy})
- The Maximum Compressive strain on the top of the subgrade (ε_{zz})

Case 3: Three layer Isotropic Elastic System



- Model all Layers as an Isotropic material, determine

- The maximum Surface Deflection and its Location

- The maximum tensile strain at the bottom of the asphalt layer (ε_{xx} or ε_{yy})

- The Maximum Compressive strain on the top of the subgrade (ε_{zz})